Mudit Gaur

Postdoctoral Scholar | The Ohio State University

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Academic Appointment

The Ohio State University — Feb. 2026 – Present
Postdoctoral Scholar, AI-EDGE Institute, Electrical and Computer Engineering — Columbus, OH

Education

Purdue University, Ph.D. in Statistics — Aug. 2019 – Dec. 2025
Advisors: Vaneet Aggarwal and Raghu Pasupathy.

Rutgers University, M.S. in Statistics — Aug. 2016 – Dec. 2017

IIT Dhanbad, Integrated M.S. in Applied Physics — Aug. 2009 – May 2014

Research Interests

My research develops statistical and optimization foundations for adaptive learning and decision-making. My interests include bilevel decision problems, inference for learned policy responses, uncertainty quantification, robust policy evaluation, and conditional generative modeling, connecting statistical precision with dependent data and finite computation.

Publications

Name in bold identifies the CV author; * denotes equal contribution.

Peer-Reviewed Journal Article

1. **Improved Sample Complexity For Diffusion Model Training Without Empirical Risk Minimizer Access**
Mudit Gaur, Prashant Trivedi, Sasidhar Kunapuli, Amrit Singh Bedi, and Vaneet Aggarwal. *Transactions on Machine Learning Research* (**TMLR**), April 2026.

Peer-Reviewed Conference Papers

1. **Order-Optimal Sample Complexity of Rectified Flows**
Hari Krishna Sahoo, **Mudit Gaur**, and Vaneet Aggarwal. **NeurIPS 2026**, accepted.
2. **Oracle-Robust Online Alignment for Large Language Models**
Zimeng Li, **Mudit Gaur**, and Vaneet Aggarwal. **NeurIPS 2026**, accepted.
3. **Discrete State Diffusion Models: A Sample Complexity Perspective**
Aadithya Srikanth*, **Mudit Gaur***, and Vaneet Aggarwal. **AISTATS 2026**.
4. **On the Sample Complexity Bounds of Bilevel Reinforcement Learning**
Mudit Gaur, Utsav Singh, Amrit Singh Bedi, Raghu Pasupathy, and Vaneet Aggarwal. **NeurIPS 2025**.
5. **Closing the Gap: Achieving Global Convergence (Last Iterate) of Actor-Critic under Markovian Sampling with Neural Network Parametrization**
Mudit Gaur, Amrit Singh Bedi, Di Wang, and Vaneet Aggarwal. **ICML 2024**; **Spotlight**.
6. **On the Global Convergence of Fitted Q-Iteration with Two-layer Neural Network Parametrization**
Mudit Gaur, Vaneet Aggarwal, and Mridul Agarwal. **ICML 2023**.

Manuscripts Under Review

1. **Conditional Diffusion Without ERM Oracles: From Finite Neural Training to Sampling Guarantees**.
   Submitted to ICLR 2027, 2026.
2. **Improved Sample Complexity Analysis for First Order Bilevel Reinforcement Learning**.
   Submitted to ICLR 2027, 2026.
3. **Policy Improvement and Preservation in Continual Reinforcement Learning with Latent Families**.
   Submitted to ICLR 2027, 2026.
4. **Constructive Softmax Training for Certified In-Context Policy Improvement**.
   Submitted to ICLR 2027, 2026.
5. **Finite-reference limits and evidence aggregation in sequential change detection**.
   Submitted to *Biometrika*, 2026.

Manuscripts in Preparation

1. **Statistical Inference for Policy-Mediated Effects in Bilevel Decision Problems**.
   Manuscript in preparation, 2026. Orthogonal estimation, identification-dependent minimax analysis, and confidence sets for response sensitivities and stationary decisions, with applications to repeated-measure intervention data.

Selected Ongoing Projects

- **Statistically calibrated robust alignment.** Robust policy improvement with finite trusted preference information and explicit uncertainty in the feedback model.
- **End-to-end neural reinforcement learning from human feedback.** Analysis connecting preference learning, reward estimation, and finite neural policy optimization.

Teaching Experience

Purdue University, Department of Statistics 2019 – 2025
Teaching Assistant West Lafayette, IN

Prepared course materials, held office hours, graded assignments, and supported students across introductory, applied, and graduate statistics courses.

Course	Title	Terms
STAT 113	Statistics and Society	Fall 2019 – Fall 2023
STAT 301	Elementary Statistical Methods	Spring 2023 – Fall 2025
STAT 511	Statistical Methods	Fall 2021
STAT 512	Applied Regression Analysis	Summer 2021
STAT 513	Introduction to Statistical Quality Control	Spring and Fall 2022; Fall 2024, 2025
STAT 514	Design of Experiments	Summer 2021, 2024, 2025
STAT 527	Introduction to Computing for Statistics	Spring and Fall 2025

Statistical Consulting

Purdue University, Statistical Consulting Service
Statistical Consultant West Lafayette, IN

- Advised more than 30 clients, including seven concurrently in Fall 2023, on research questions and statistical analyses.
- Implemented quantile regression, clustering, and time-series models; provided software support for statistical analyses.
- Staffed a software help desk providing coding assistance in Python, R, SAS, and SPSS.
- Invited to coauthor a client's research project in recognition of consulting contributions.

Industry Experience

Encore Capital Group Feb. 2018 – July 2019
Statistician San Diego, CA

- Developed, maintained, and deployed statistical and machine learning models to support collections decisions and reduce payment attrition.
- Identified payer groups at risk of discontinuing payments in the legal collections channel.
- Analyzed spatial and temporal patterns in annual collections to characterize geographic variation.

Honors and Scholarships

B & M McLean Graduate Scholarship, Purdue University, Department of Statistics. Awarded on the basis of academic merit.

INSPIRE Scholarship, Government of India. Supported study in the basic sciences.

Technical Skills

Programming and statistical software: Python, R, SAS, SPSS.

References

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Amrit Singh Bedi
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